Rahul G

gilakarahul@gmail.com | Portfolio | Github | Linkedin

SUMMARY

MACHINE LEARNING ENGINEER
Engineer with 2 years of experience building end-to-end ML systems from data pipelines to deployment and monitoring. Experienced in FastAPI, TensorFlow, PyTorch, CI/CD, Langfuse observability, and production applications in computer vision, NLP, and generative AI.

EXPERIENCE

BRISKBE PVT LTD (TRYCOM.AI) | Software Developer (AI/ML)
Jul 2024 -- Oct 2025 | Chennai

- Built an agent-based SEO topic clustering and LLM content generation pipeline using LangChain, LangGraph, and Qdrant, automating blog creation workflows.
- Developed an async OCR and LLM-based image generation pipeline, reducing content creation time from 20–30 minutes to under 3 minutes.
- Architected Langfuse-based observability for granular LLM cost tracking across users and third-party APIs.
- Developed and maintained production-grade FastAPI services powering AI automation workflows.

MOCERO HEALTH SOLUTIONS | Machine Learning Intern
Jan 2024 -- Jul 2024 | Chennai

- Built a multilingual image localization pipeline enabling Generative AI images with native Tamil text using PaddleOCR, IndicTrans2, LaMa, and OpenCV.
- Developed an image scraping and CLIP-based ranking system for large-scale visual dataset curation.
- Automated infographic generation using template rendering and LLM-generated content.

CAMEO CORPORATE SERVICES | Machine Learning Intern
Sep 2021 -- Dec 2021 | Chennai

- Developed a TensorFlow-based KYC document classifier covering 20 document types with 83% accuracy.
- Reduced manual tagging effort by 50% through automated inference pipelines.
- Integrated prediction outputs with Oracle Database to streamline downstream processing.

PROJECTS

TRANSISTOR DEFECT INSPECTION | PatchCore, WideResNet50, FastAPI, OpenCV, Docker | Github

- Developed an unsupervised defect detection system using PatchCore with WideResNet50 for transistor inspection, achieving strong image-level classification performance.
- Built an inference pipeline that generated anomaly heatmaps for precise pixel localization of defects on uploaded images.
- Containerized and deployed the end-to-end application using FastAPI and Docker for production-ready inference.

PUBLICATION

IEEE ICCSP 2024 - Chronic Obstructive Pulmonary Disease Severity Classification using Lung Sound IEEE

SKILLS

PROGRAMMING
- Python
- C/C++
- SQL

AI / ML
- **Frameworks** - TensorFlow, PyTorch
- **LLM** - GPT-4, Qwen, RAG
- **Data Analysis** - NumPy, Pandas, Matplotlib, OpenCV
- **Anomaly Detection** using PatchCore
- **Object Detection & Segmentation** using YOLO
- **Classification & Clustering** using Scikit-learn
- **Vision Language Model** - CLIP , SigLIP
- **OCR** using Keras-OCR , PaddleOCR

BACKEND & TOOLS
- **API Development** with FastAPI
- **Containerization** using Docker
- **CI/CD** with Git and GitHub Actions
- **Databases** - MongoDB, MySQL, Qdrant
- **Task Queue** - Celery + Redis
- **Testing** using PyTest
- **LLM Monitoring** using Langfuse

ACHIEVEMENTS

- **Best Sustainable Hack** - NIT Bhopal Version Beta Hackathon
- **Top Participant** - HackFest by SirionLabs
- **Semi Finalist** - Smart India Hackathon

CERTIFICATIONS

- **IIT-Madras** - Object Detection and Image Segmentation
- **IIT-Madras** - Make your own DALL-E
- **AWS** - AI Practitioner Learning Plan
- **AWS** - Cloud Practitioner Essentials

EDUCATION

SRI VENKATESWARA COLLEGE OF ENGINEERING
B.Tech in Artificial Intelligence and Data Science
2020 - 2024 | CGPA: 9.07 | Chennai